*Photographs by* BANDEEP SINGH



# LI-ION, KING OF THE E-WASTE JUNGLE

## LITHIUM BATTERY RECYCLING

- With the rise in electric vehicles, Lithium-ion (Li-ion) battery recycling has become the next big thing
- Recyclers like Attero and Exigo are among the early entrants in India
- Attero plans to grow plant capacity from 3,500 metric tonnes to 16,000 tonnes in a year
- Exigo is setting up new 7,500 metric tonne plant

## THE PROCESS

- Li-ion batteries come with different chemistries
- EV batteries commonly have nickel-manganese-cobalt as their cathode and graphite as the anode. In between are separators made of aluminium and copper foil
- There are three broad approaches to recycling Li-ion batteries
- Battery is processed mechanically to get two outputs: iron/copper/aluminium flakes and black mass. The last is a mixture of electrode materials—cobalt, lithium, nickel, graphite and manganese
- The second approach is via smelting to extract these materials (not used in India)
- Hydrometallurgy (used by Attero) using chemical technologies like leaching and solvent extraction to extract electrode materials constitutes the third approach

Recycling companies globally use these methods in various hybrid combinations